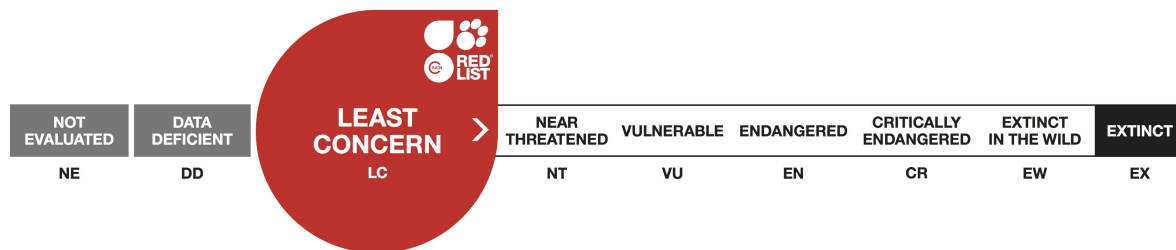


Notoryctes typhlops, Itjaritjari

Assessment by: Burbidge, A.A. & Woinarski, J.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Notoryctemorphia	Notoryctidae

Taxon Name: *Notoryctes typhlops* (Stirling, 1889)

Synonym(s):

- *Psammoryctes typhlops* Stirling, 1889

Common Name(s):

- English: Itjaritjari, Marsupial Mole, Southern Marsupial Mole
- French: Grande taupe marsupiale

Assessment Information

Red List Category & Criteria: Least Concern [ver 3.1](#)

Year Published: 2016

Date Assessed: March 15, 2014

Justification:

Evaluated as Least Concern because the Itjaritjari is widespread and there is no evidence of a continuing decline.

Previously Published Red List Assessments

2008 – Data Deficient (DD) – <http://dx.doi.org/10.2305/IUCN.UK.2008.RLTS.T14879A4467905.en>

1996 – Endangered (EN)

Geographic Range

Range Description:

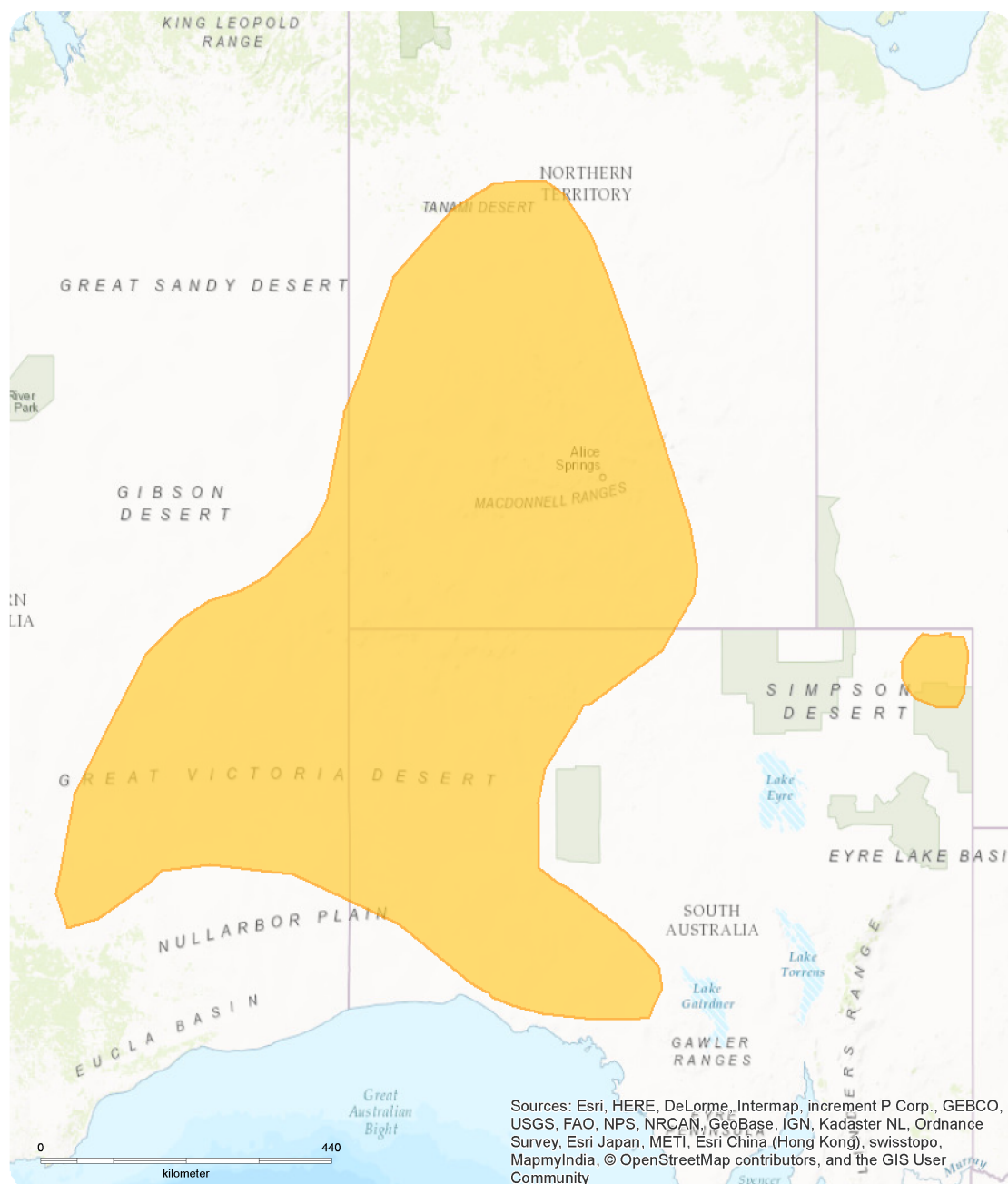
Itjaritjari (Southern Marsupial Mole) occur in the Great Victoria Deserts of Western Australia and South Australia and the western half of the Simpson Desert in southern Northern Territory and Queensland. Their distribution follows that of the sand dune habitat that they prefer. The Itjaritjari may be sympatric with Kakarratul (*Notoryctes caurinus*) in northern parts of the former's range.

Country Occurrence:

Native: Australia (Northern Territory, Queensland, South Australia, Western Australia)

Distribution Map

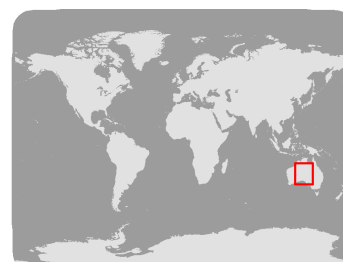
Notoryctes typhlops



Range

Extant (resident)

Compiled by:
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Conservation of Nature)



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.



Population

There is no robust estimate of population size, nor any data that suggest decline. However, evidence can be found in most areas of suitable habitat (stable dunes) within the species' distribution. Indices of levels of activity have been developed, but these are not yet correlated with abundance.

Current Population Trend: Stable

Habitat and Ecology (see Appendix for additional information)

Itjaritjari inhabit sand dunes and adjacent swales where there is suitable deep, loose sand. They spend almost their entire lives underground, only very occasionally coming to the surface, and only remaining on the surface for a short time. Most specimens and sightings come from animals found on the surface. However, it appears they rarely come to the surface, which makes them less prone to predation by Red Foxes, Cats and other predators.

When underground, marsupial moles virtually swim through the soil. The sand is packed in behind the animal as it progresses and no empty tunnel is formed. Nevertheless, where one has passed through a soil profile, an oval shaped differentiation in soil texture and colour can be seen (Johnson and Walton 1989). Where Itjaritjari are known to occur, their underground backfilled tunnels are often very common and these may represent an important source of soil disturbance (biopedturbation) (Whitford and Kay 1999). Evidence of 30-60 kilometres of backfilled tunnel per hectare is usual in these areas (Benshemesh 2004), representing about 40-80 m³ of turned-over soil per hectare, or up to 1% of the soil within one metre of the surface.

Itjaritjari are insectivorous, capturing their prey while underground. Examination of gut contents of preserved specimens and of the habitat they occupy, has revealed a generalist diet comprising two main prey types: social insects (ants and termites) and the larvae of beetles. Ants, termites and beetle larvae were also the main invertebrates captured in soil cores on sandridges; other invertebrates combined contributed <5% to abundance (Pavey *et al.* 2012).

Recent studies suggest that marsupial moles may be common prey items of larger mammalian predators. Paltridge (1998) found remains of marsupial moles in predator scats at five of her six study sites in the Tanami Desert. Her study revealed that 10% of Red Fox scats, 3% of Cat scats and 5% of Dingo scats contained signs of moles (n= 82, 111 and 59 respectively). Signs of marsupial moles have also been found in Red Fox and Dingo scats from the Anangu-Pitjantjatjara / Yankunytjatjara Lands in South Australia (Benshemesh 2004), and in other areas of the Northern Territory (Paltridge 1998), although not as frequently. Benshemesh *et al.* (2010) examined nearly 1800 predator scats from both southern and northern dunefields and found little difference between the likelihood of finding marsupial mole signs in scats of Red Foxes, cats and Dingoes, but major differences between northern and southern areas. In southern areas inhabited by Itjaritjari, only 1% of predator scats contained traces of marsupial moles, compared with 8% in northern areas largely inhabited by Kakarratul, a difference they attributed to the higher abundance of marsupial moles in the north and shallower burrowing habits. It is uncertain whether these predators take marsupial moles on the surface or dig them up, or indeed whether they are actually killing moles or taking dead animals. Dead or severely debilitated moles have been recorded on the surface on several occasions (Benshemesh 2004), but it seems most likely that these predators prey upon living moles that are on or just under the surface. R. T. Maurice,

who travelled widely in the Great Victoria Desert between 1897 and 1903, reported that local Aboriginal people were able to capture Itjaritjari after hearing them when they were under the surface (Gara 1996 cited in Benshemesh 2004), and larger mammalian predators may do likewise. Benshemesh *et al.* (2010) found some evidence that Dingoes, at least, hunted marsupial moles in northern areas, where it was common to find the remains of more than one marsupial mole in each scat. On the surface, marsupial moles are vulnerable to a wide range of predators including birds of prey, snakes and goannas.

A study of the metabolic physiology of a single Kakarratul concluded that it differs in some aspects of thermal and metabolic physiology from other marsupials, most likely reflecting its almost completely fossorial existence (Withers *et al.* 2000). Itjaritjari are likely to be similar.

Systems: Terrestrial

Threats (see Appendix for additional information)

While several possible threats have been identified (predation by introduced species, inappropriate fire regimes), recent data suggests Itjaritjari remain widespread and relatively abundant.

Credits

Assessor(s): Burbidge, A.A. & Woinarski, J.

Reviewer(s): Hawkins, C.

Contributor(s): Benshemesh, J., Copley, P., Johnson, K., McGilvray, A. & Ward, S.

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External Resources

For [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
3. Shrubland -> 3.5. Shrubland - Subtropical/Tropical Dry	Resident	Suitable	Yes
4. Grassland -> 4.5. Grassland - Subtropical/Tropical Dry	Resident	Suitable	Yes
8. Desert -> 8.1. Desert - Hot	Resident	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
11. Climate change & severe weather -> 11.2. Droughts	Ongoing	Whole (>90%)	Negligible declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance		
11. Climate change & severe weather -> 11.3. Temperature extremes	Ongoing	Whole (>90%)	Negligible declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance		
2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.4. Scale Unknown/Unrecorded	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.1. Fire & fire suppression -> 7.1.3. Trend Unknown/Unrecorded	Ongoing	Whole (>90%)	Negligible declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.1. Unspecified species	Ongoing	Whole (>90%)	Negligible declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.4. Problematic species/disease of unknown origin -> 8.4.2. Named species (<i>Felis catus</i>)	Ongoing	-	-	-
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.4. Problematic species/disease of unknown origin -> 8.4.2. Named species (<i>Vulpes vulpes</i>)	Ongoing	-	-	-
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

8. Invasive and other problematic species, genes & diseases -> 8.4. Problematic species/disease of unknown origin -> 8.4.2. Named species (Canis lupus ssp. dingo)	Ongoing	-	-	-
Stresses: 2. Species Stresses -> 2.1. Species mortality				

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions in Place
In-Place Research, Monitoring and Planning
Action Recovery plan: Yes
Systematic monitoring scheme: No
In-Place Land/Water Protection and Management
Conservation sites identified: Yes, over part of range
Occur in at least one PA: Yes
Percentage of population protected by PAs (0-100): 1-10
Area based regional management plan: No
Invasive species control or prevention: Yes
In-Place Species Management
Harvest management plan: No
Successfully reintroduced or introduced benignly: No
Subject to ex-situ conservation: No
In-Place Education
Subject to recent education and awareness programmes: Yes
Included in international legislation: Yes
Subject to any international management/trade controls: Unknown

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions Needed
1. Land/water protection -> 1.1. Site/area protection
1. Land/water protection -> 1.2. Resource & habitat protection
2. Land/water management -> 2.1. Site/area management
2. Land/water management -> 2.2. Invasive/problematic species control

Conservation Actions Needed
2. Land/water management -> 2.3. Habitat & natural process restoration
4. Education & awareness -> 4.3. Awareness & communications

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.1. Taxonomy
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.5. Threats
1. Research -> 1.6. Actions
2. Conservation Planning -> 2.2. Area-based Management Plan
3. Monitoring -> 3.1. Population trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 20000-40000
Continuing decline in area of occupancy (AOO): No
Extreme fluctuations in area of occupancy (AOO): No
Estimated extent of occurrence (EOO) (km ²): 620000
Continuing decline in extent of occurrence (EOO): No
Extreme fluctuations in extent of occurrence (EOO): No
Number of Locations: 10-50
Population
Number of mature individuals: 10000-100000
Continuing decline of mature individuals: No
Extreme fluctuations: No
Population severely fragmented: No
Continuing decline in subpopulations: No
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No

Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: No
Generation Length (years): 10-15
Movement patterns: Not a Migrant

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